

Appendix A: Golden Benchmark Suite

(50-Query Dataset)

The following dataset represents the empirical ground truth used to evaluate the spatial search architecture. It consists of 50 manually curated natural language queries mapped to their executable PostGIS Gold SQL counterparts, stratified by semantic cluster and difficulty level.

A.1 Food and Dining Logistics

[01] [Easy] Where can I get coffee?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND amenity ILIKE '%cafe%' ORDER BY name ASC;
```

[02] [Easy] Find all ice cream parlors.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND amenity ILIKE '%ice_cream%' ORDER BY name ASC;
```

[03] [Easy] Show me all restaurants.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND amenity ILIKE '%restaurant%' ORDER BY name ASC;
```

[04] [Easy] List all fast food places.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND amenity ILIKE '%fast_food%' ORDER BY name ASC;
```

[05] [Easy] Find bakeries near campus.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND (shop ILIKE '%bakery%' OR amenity ILIKE '%bakery%') ORDER BY name ASC;
```

[06] [Medium] Find me restaurants or fast food places.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(amenity ILIKE '%restaurant%' OR amenity ILIKE '%fast_food%') ORDER BY name ASC;
```

[07] [Medium] Show me places where I can get takeaway food.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
takeaway ILIKE '%yes%' ORDER BY name ASC;
```

[08] [Medium] Find places with outdoor seating.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
outdoor_seating ILIKE '%yes%' ORDER BY name ASC;
```

[09] [Hard] Find cafes or restaurants open 24/7.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(amenity ILIKE '%cafe%' OR amenity ILIKE '%restaurant%') AND opening_hours ILIKE  
'%24/7%' ORDER BY name ASC;
```

[10] [Hard] Show me non-vegetarian food places that are cafes.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
amenity ILIKE '%cafe%' AND "diet:non-vegetarian" ILIKE '%yes%' ORDER BY name ASC;
```

A.2 Sports and Recreation

[11] [Easy] Where can I play badminton?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
sport ILIKE '%badminton%' ORDER BY name ASC;
```

[12] [Easy] Where is the swimming pool?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(leisure ILIKE '%swimming_pool%' OR amenity ILIKE '%swimming_pool%') ORDER BY name  
ASC;
```

[13] [Easy] Where can I play cricket?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(sport ILIKE '%cricket%' OR amenity ILIKE '%cricket%') ORDER BY name ASC;
```

[14] [Medium] Show me all sports pitches and courts.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(leisure ILIKE '%pitch%' OR leisure ILIKE '%sports_centre%') ORDER BY name ASC;
```

[15] [Medium] Find volleyball or table tennis courts.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(sport ILIKE '%volleyball%' OR sport ILIKE '%table_tennis%') ORDER BY name ASC;
```

[16] [Medium] Find parks or gardens to relax in.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(leisure ILIKE '%park%' OR leisure ILIKE '%garden%') ORDER BY name ASC;
```

[17] [Hard] Find lit sports pitches with grass surface.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
leisure ILIKE '%pitch%' AND surface ILIKE '%grass%' AND lit ILIKE '%yes%' ORDER BY name  
ASC;
```

[18] [Hard] Show me indoor sports facilities.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(leisure ILIKE '%sports_centre%' OR sport IS NOT NULL) AND indoor ILIKE '%yes%' ORDER  
BY name ASC;
```

A.3 Residential Infrastructure

[19] [Easy] List all dormitories on campus.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
building ILIKE '%dormitory%' ORDER BY name ASC;
```

[20] [Easy] Show me residential buildings.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
building ILIKE '%residential%' ORDER BY name ASC;
```

[21] [Medium] Find all bhawans or hostels.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(building ILIKE '%dormitory%' OR name ILIKE '%bhawan%' OR name ILIKE '%hostel%')  
ORDER BY name ASC;
```

[22] [Medium] Find apartments or residential complexes.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(building ILIKE '%apartments%' OR building ILIKE '%residential%') ORDER BY name ASC;
```

[23] [Hard] Find Meera Bhawan girls hostel.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(name ILIKE '%Meera%' OR name ILIKE '%Bhawan%' OR name ILIKE '%hostel%') ORDER BY  
name ASC;
```

[24] [Hard] Find Ram Bhawan or Krishna Bhawan.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(name ILIKE '%Ram Bhawan%' OR name ILIKE '%Krishna Bhawan%') ORDER BY name ASC;
```

[25] [Extra] Find Srinivasa Ramanujan Bhawan boys hostel.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(name ILIKE '%Srinivasa Ramanujan%' OR name ILIKE '%boys hostel%') ORDER BY name  
ASC;
```

[26] [Extra] Show me Malviya Bhawan block A and block B.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(name ILIKE '%Malviya Bhawan Block A%' OR name ILIKE '%Malviya Bhawan Block B%')  
ORDER BY name ASC;
```

A.4 Academics and Education

[27] [Easy] Where is the library?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
amenity ILIKE '%library%' ORDER BY name ASC;
```

[28] [Easy] Find the university.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(amenity ILIKE '%university%' OR building ILIKE '%university%' OR name ILIKE '%bits%')  
ORDER BY name ASC;
```

[29] [Easy] Where are the schools on campus?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(amenity ILIKE '%school%' OR education ILIKE '%school%') ORDER BY name ASC;
```

[30] [Medium] Show me university department buildings.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(building ILIKE '%university=department%' OR building ILIKE '%university%department%')  
ORDER BY name ASC;
```

[31] [Medium] Find colleges or research institutes.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(amenity ILIKE '%college%' OR amenity ILIKE '%research_institute%') ORDER BY name ASC;
```

[32] [Hard] Find university buildings that are not places of worship.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
building ILIKE '%university%' AND (amenity IS NULL OR amenity NOT ILIKE  
'%place_of_worship%') ORDER BY name ASC;
```

[33] [Hard] Find the research institute or conference centre.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(amenity ILIKE '%research_institute%' OR amenity ILIKE '%conference_centre%') ORDER BY  
name ASC;
```

A.5 Public Utilities and Logistics

[34] [Easy] Where is the post office?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
amenity ILIKE '%post_office%' ORDER BY name ASC;
```

[35] [Easy] Find the fuel station.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND amenity ILIKE '%fuel%' ORDER BY name ASC;
```

[36] [Medium] Which places accept UPI payment?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND "payment:upi" = 'yes' ORDER BY name ASC;
```

[37] [Medium] Show me places that accept Google Pay.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND "payment:google_pay" = 'yes' ORDER BY name ASC;
```

[38] [Medium] Find places with wifi or internet access.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND (internet_access ILIKE '%wlan%' OR internet_access ILIKE '%yes%') ORDER BY name ASC;
```

[39] [Hard] Show me food places with UPI or Google Pay that accept takeaway.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND (amenity ILIKE '%cafe%' OR amenity ILIKE '%restaurant%' OR amenity ILIKE '%fast_food%') AND ("payment:upi" = 'yes' OR "payment:google_pay" = 'yes') AND takeaway ILIKE '%yes%' ORDER BY name ASC;
```

[40] [Hard] Find government offices or courthouse.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND (amenity ILIKE '%courthouse%' OR office ILIKE '%government%') ORDER BY name ASC;
```

A.6 Healthcare Services

[41] [Easy] Where are the hospitals?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND amenity ILIKE '%hospital%' ORDER BY name ASC;
```

[42] [Easy] Find the pharmacy.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND amenity ILIKE '%pharmacy%' ORDER BY name ASC;
```

[43] [Medium] Where are the hospitals or clinics?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND (amenity ILIKE '%hospital%' OR amenity ILIKE '%clinic%') ORDER BY name ASC;
```

[44] [Medium] Find dentists or healthcare facilities.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND (amenity ILIKE '%dentist%' OR healthcare IS NOT NULL) ORDER BY name ASC;
```

[45] [Hard] Find a blood bank or hospital with a phone number.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND (amenity ILIKE '%blood_bank%' OR amenity ILIKE '%hospital%') AND phone IS NOT NULL ORDER BY name ASC;
```

A.7 Vibe, Culture, and Tourism

[46] [Easy] Where is the museum?

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND tourism ILIKE '%museum%' ORDER BY name ASC;
```


[47] [Easy] Find all statues on campus.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
artwork_type ILIKE '%statue%' ORDER BY name ASC;
```

[48] [Medium] Show me historic places like memorials or manors.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(historic ILIKE '%memorial%' OR historic ILIKE '%manor%') ORDER BY name ASC;
```

[49] [Hard] Show me photogenic spots like statues, memorials or museums.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
(tourism ILIKE '%museum%' OR tourism ILIKE '%artwork%' OR historic ILIKE '%memorial%' OR  
artwork_type ILIKE '%statue%') ORDER BY name ASC;
```

[50] [Extra] Find the Birla aircraft or aircraft memorial.

SQL:-

```
SELECT name FROM pilani_places WHERE name IS NOT NULL AND name != 'nan' AND  
name ILIKE '%Birla%' AND (historic ILIKE '%aircraft%' OR historic ILIKE '%memorial%' OR  
name ILIKE '%aircraft%') ORDER BY name ASC;
```